



Conceptos básicos de Electrónica Digital



Autor: Ing. Marcelo Casasnovas mail: marcelocasasnovas@gmail.com



Alimentación de circuitos simbología



Baterías

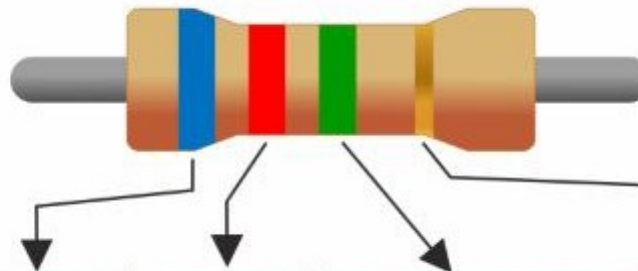


Nodos de voltajes



Componentes básicos

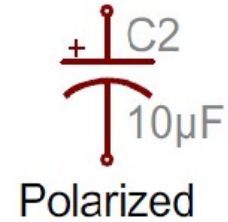
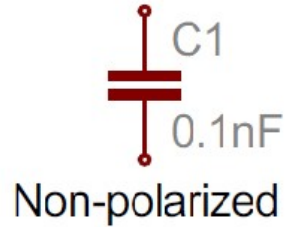
Resistencias



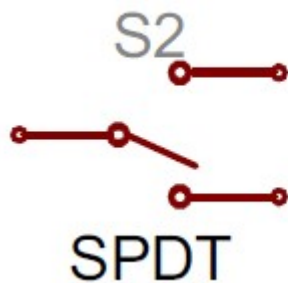
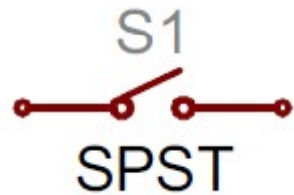
COLOR	BANDA 1	BANDA 2	MULTIPLICADOR	TOLERANCIA
NEGRO	0	0	$\times 1 \Omega$	
MARRÓN	1	1	$\times 10 \Omega$	+ / - 1%
ROJO	2	2	$\times 100 \Omega$	+ / - 2%
NARANJA	3	3	$\times 1000 \Omega$	
AMARILLO	4	4	$\times 10,000 \Omega$	
VERDE	5	5	$\times 100,000 \Omega$	
AZUL	6	6	$\times 1,000,000 \Omega$	
VIOLETA	7	7	$\times 10,000,000 \Omega$	
GRIS	8	8	$\times 100,000,000 \Omega$	
BLANCO	9	9	$\times 1,000,000,000 \Omega$	
DORADO			$\times 0,1 \Omega$	+ / - 5%
PLATEADO			$\times 0,01 \Omega$	+ / - 10%



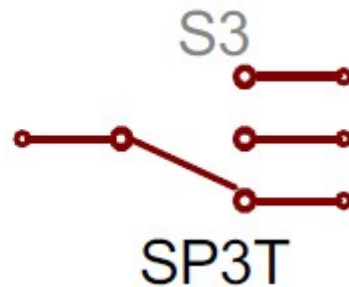
Componentes básicos



Capacitores

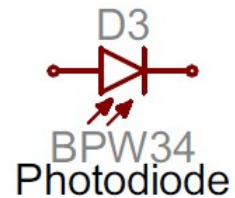
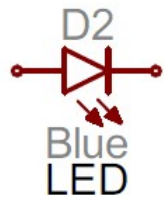


Interruptores

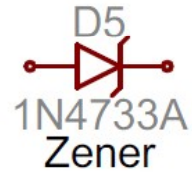




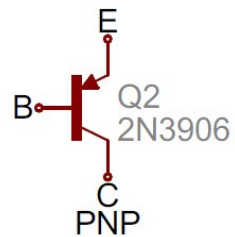
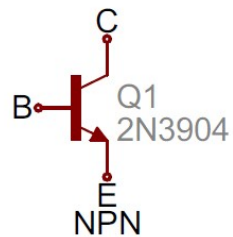
Componentes básicos



Diodos Led



Diodos

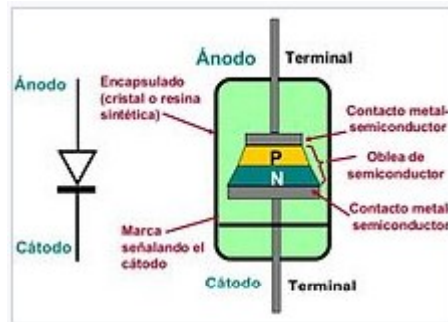


Transistores

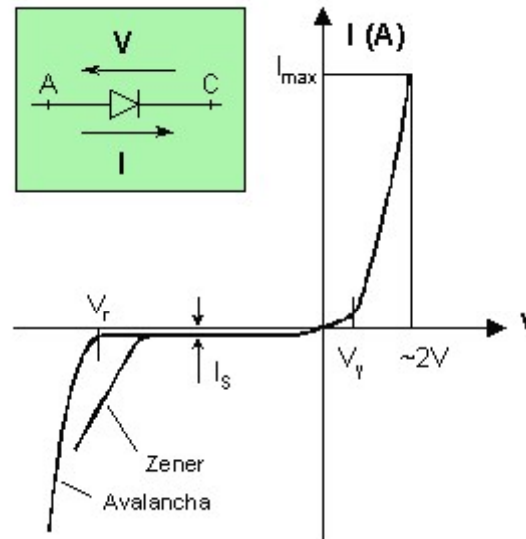


Componentes básicos

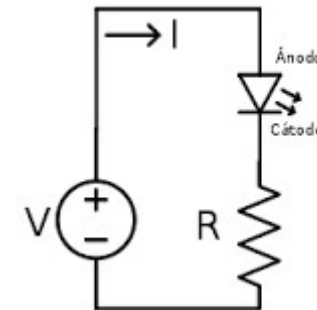
Polarización de Diodos y Leds



Terminales de un Diodo



Curva característica de un Diodo

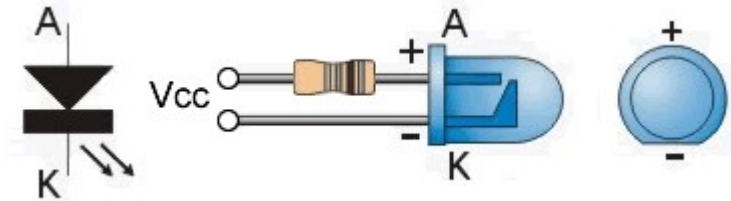
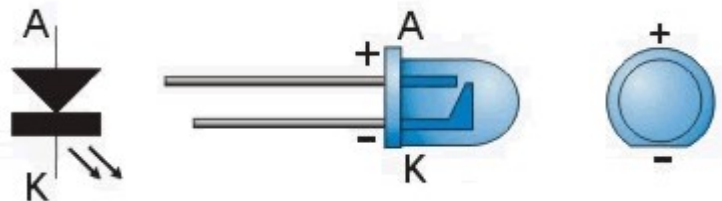


Conexión básica



Componentes básicos

Polarización de Diodos Led



Led con resistencia limitadora de corriente

Valores de resistencia en función de la tensión de alimentación

Alimentación: 12V			
tipo de led	Vled	corriente	resistencia
 azul / blanco alta luminosidad	3,7V	20 mA	(calculado: 415 ohm)  390 ohm
 rojo alta luminosidad	1,2V	20 mA	(calculado: 540 ohm)  560 ohm
 rojo tipo indicatore	1,2V	5 mA	(calculado: 2160 ohm)  2200 ohm
 verde / amarillo tipo indicatore	1,6V	5 mA	(calculado: 2080 ohm)  2200 ohm

Alimentación: 5V			
tipo de led	Vled	corriente	resistencia
 azul / blanco alta luminosidad	3,7V	20 mA	(calculado: 65 ohm)  68 ohm
 rojo alta luminosidad	1,2V	20 mA	(calculado: 190 ohm)  180 ohm
 rojo tipo indicatore	1,2V	5 mA	(calculado: 760 ohm)  680 ohm
 verde / amarillo tipo indicatore	1,6V	5 mA	(calculado: 680 ohm)  680 ohm

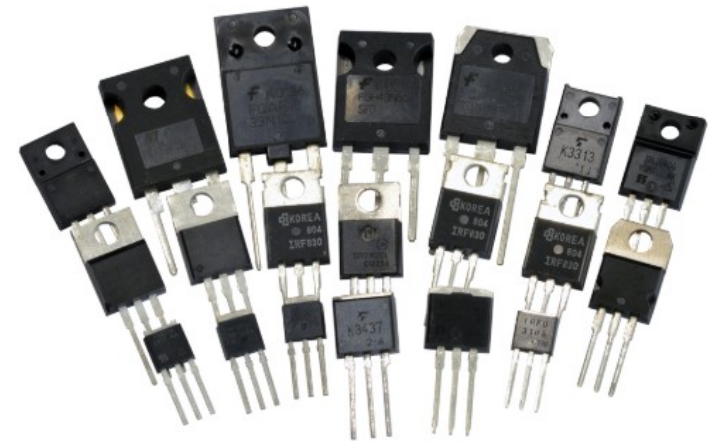
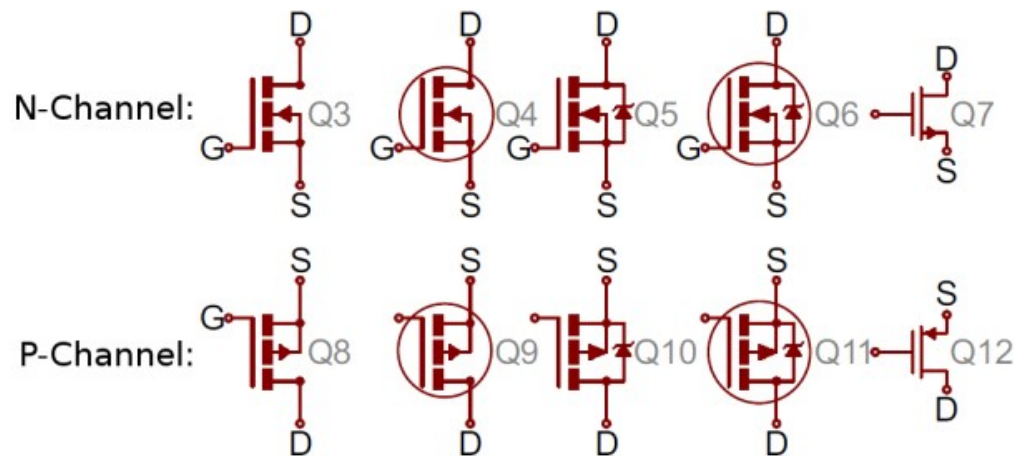
Led en funcionamiento





Componentes básicos

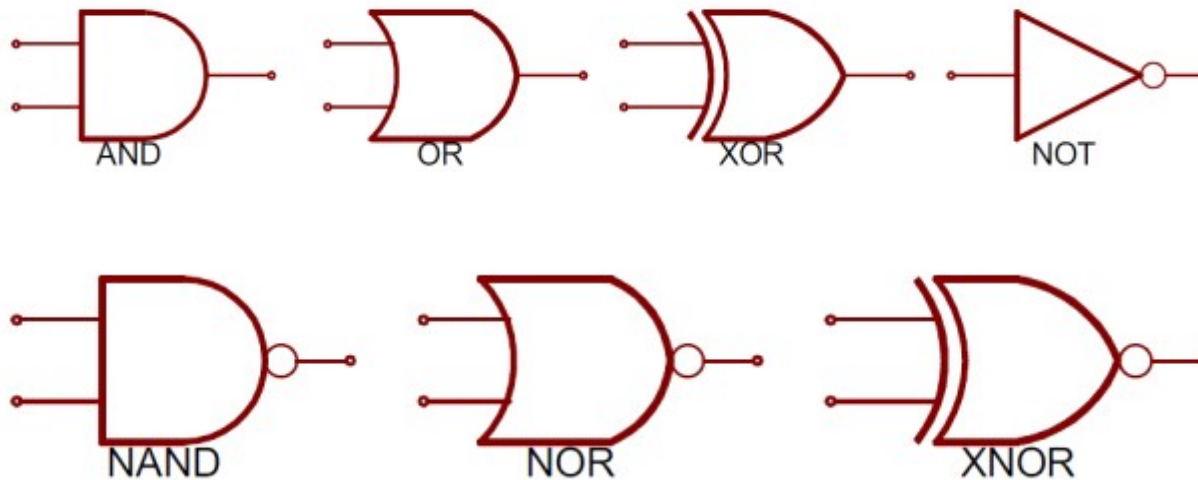
Transistores Mosfet





Componentes básicos

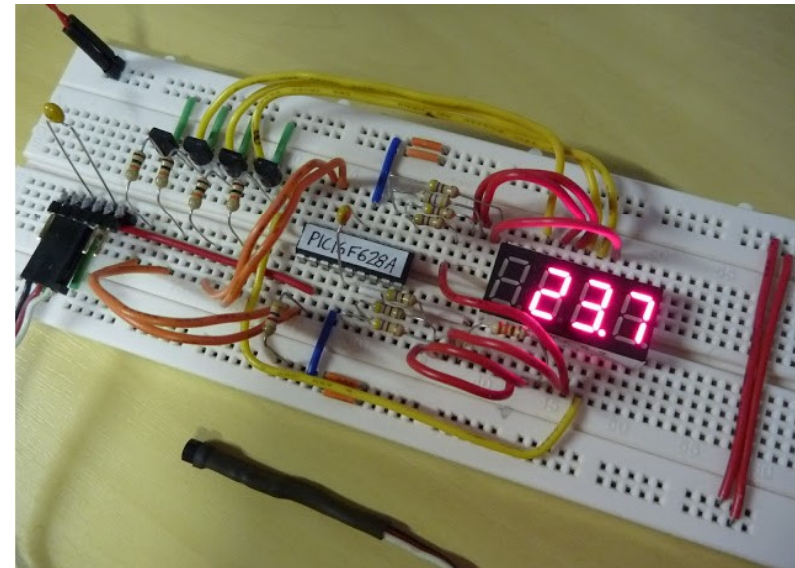
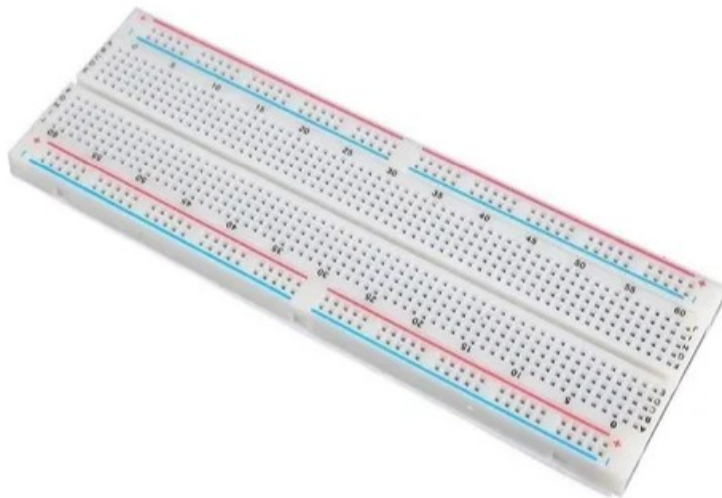
Compuertas Lógicas





Componentes básicos

Protoboard





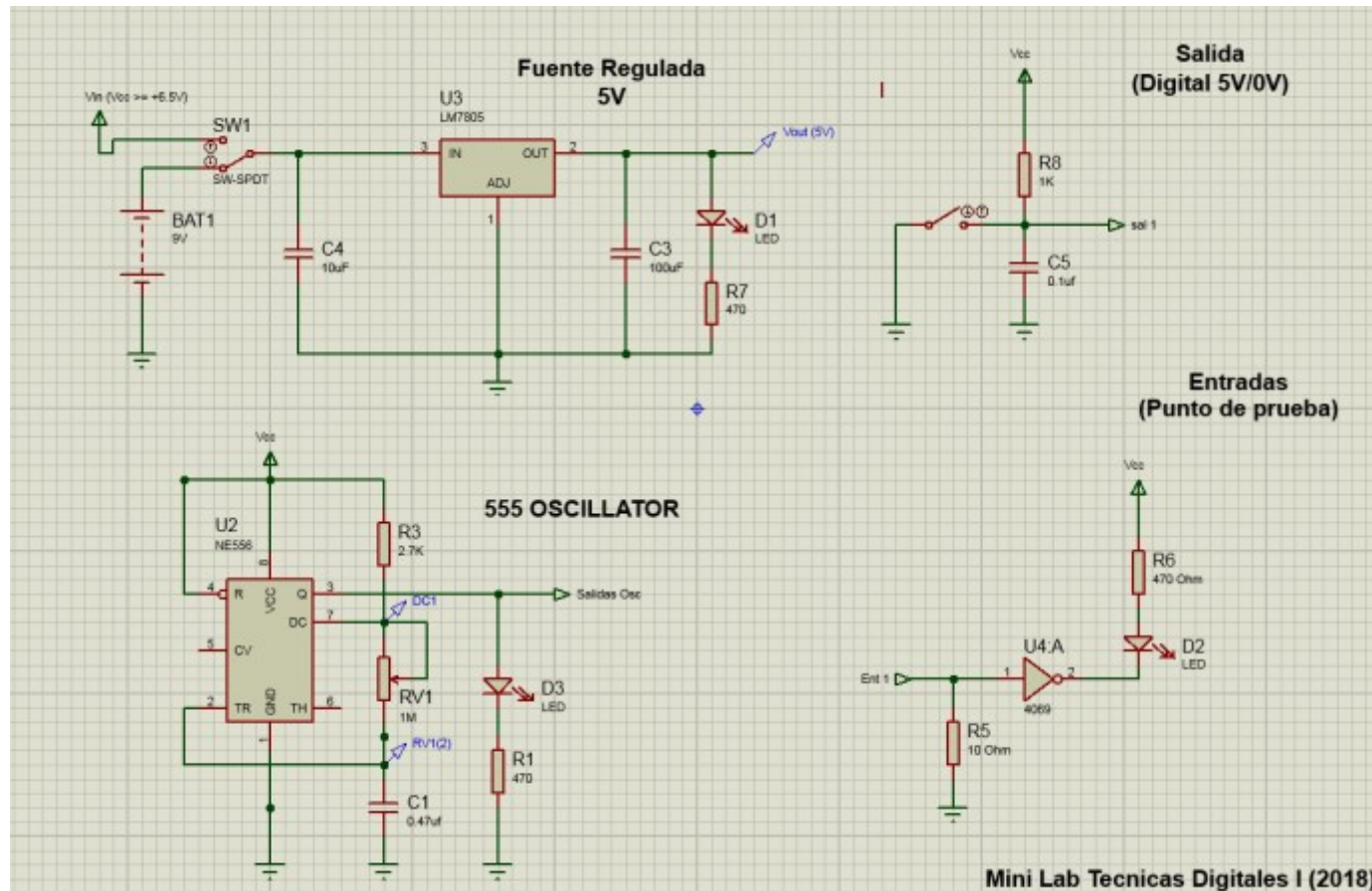
Herramientas



- Soldador
- Multímetro
- Estaño
- Pinzas
- Alicates
- Desoldador

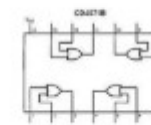
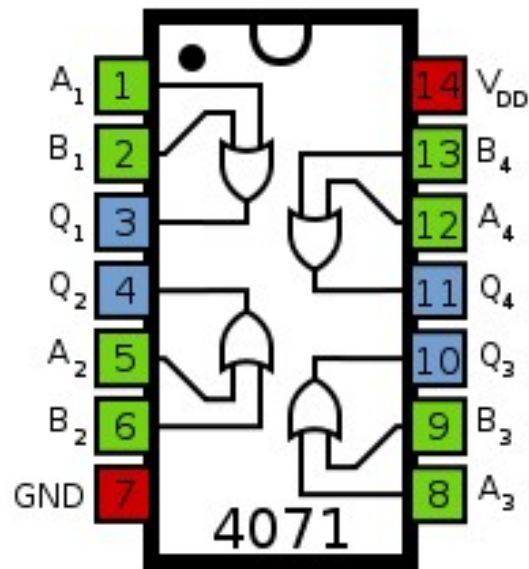


MINILAB





Circuito integrado CD4071





Hoja de datos

National Semiconductor February 1988

CD4071BM/CD4071BC
Quad 2-Input OR Buffered B Series Gate
CD4081BM/CD4081BC
Quad 2-Input AND Buffered B Series Gate

General Description
These quad gates are monolithic complementary MOS (CMOS) integrated circuits constructed with N- and P-channel enhancement mode transistors. They have equal source and sink current capabilities and conform to standard B series output drive. The devices also have buffered outputs which improve transfer characteristics by providing very high gain.
All inputs protected against static discharge with diodes to V_{DD} and V_{SS} .

Features

- Low power TTL compatibility
- Fan out of 2 driving 74L or 1 driving 74LS
- 5V–10V–15V parametric ratings
- Symmetrical output characteristics
- Maximum input leakage 1 μA at 15V over full temperature range

Connection Diagrams

CD4071B Dual-In-Line Package

Top View TL/F/5877-3

CD4081B Dual-In-Line Package

Top View TL/F/5877-6

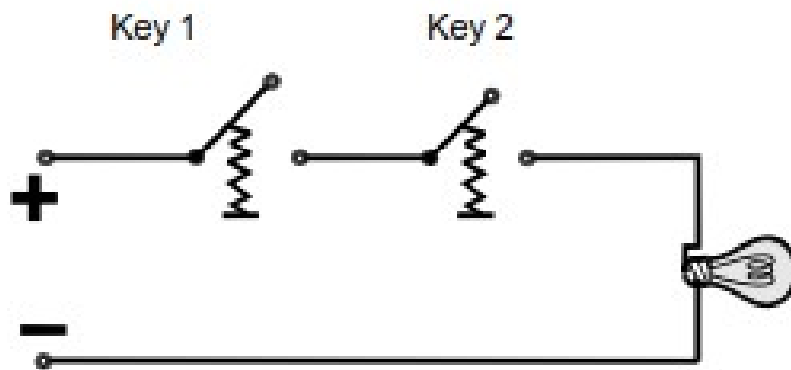
Order Number CD4071B or CD4081B

CD4071BM/CD4071BC Quad 2-Input OR Buffered B Series Gate
CD4081BM/CD4081BC Quad 2-Input AND Buffered B Series Gate



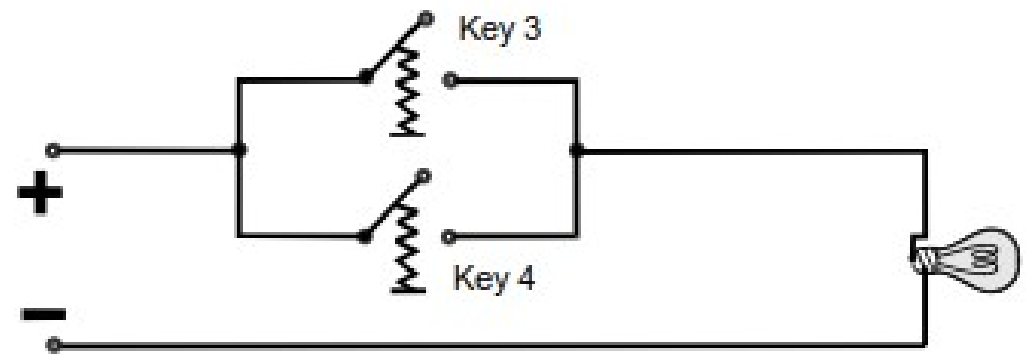
Tipos de conexiones de circuitos

Circuito serie



Función **AND**

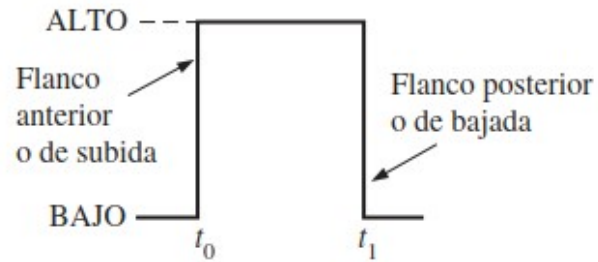
Circuito paralelo



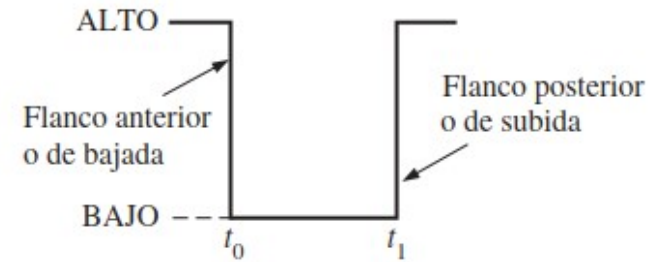
Función **OR**



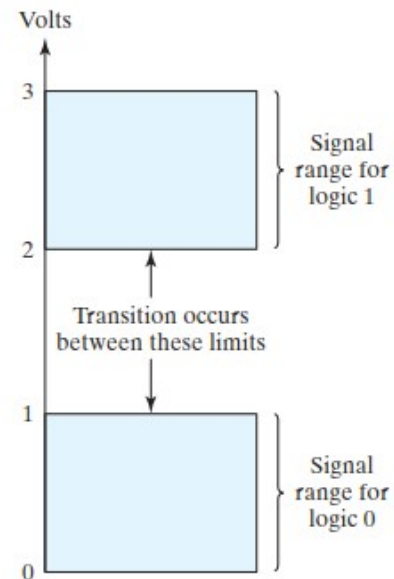
Formas de ondas Digitales



(a) Impulso positivo



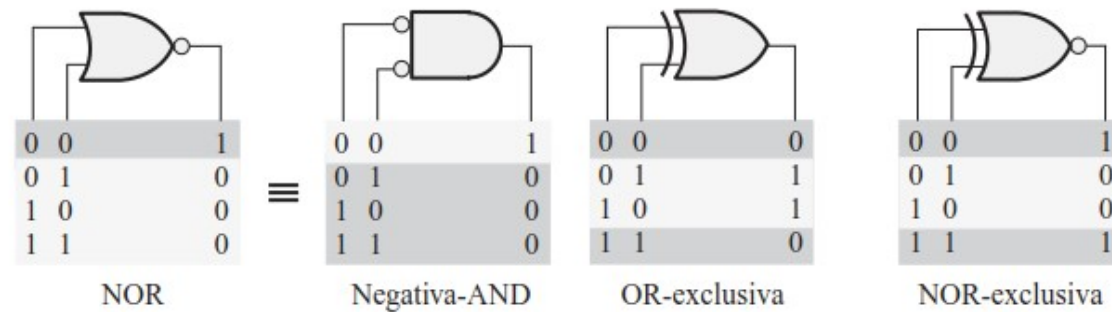
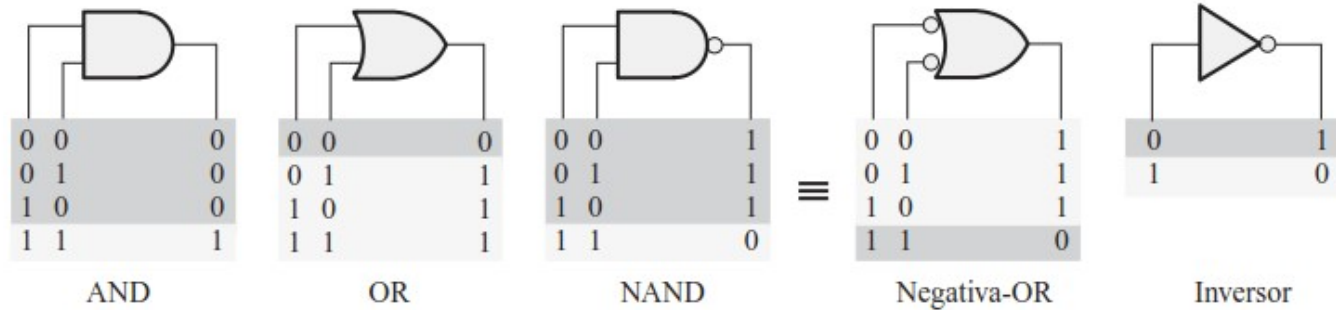
(b) Impulso negativo



Rangos de niveles lógicos



Compuertas Lógicas



(a) Two-input AND gate



(b) Two-input OR gate



(c) NOT gate or inverter



Compuertas Lógicas

AND



$$F = x \cdot y$$

x	y	F
0	0	0
0	1	0
1	0	0
1	1	1

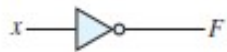
OR



$$F = x + y$$

x	y	F
0	0	0
0	1	1
1	0	1
1	1	1

Inverter



$$F = x'$$

x	F
0	1
1	0

Buffer



$$F = x$$

x	F
0	0
1	1

NAND



$$F = (xy)'$$

x	y	F
0	0	1
0	1	1
1	0	1
1	1	0

NOR



$$F = (x + y)'$$

x	y	F
0	0	1
0	1	0
1	0	0
1	1	0

Exclusive-OR
(XOR)



$$F = xy' + x'y$$
$$= x \oplus y$$

x	y	F
0	0	0
0	1	1
1	0	1
1	1	0

Exclusive-NOR
or
equivalence



$$F = xy + x'y'$$
$$= (x \oplus y)'$$

x	y	F
0	0	1
0	1	0
1	0	0
1	1	1



Compuertas Lógicas

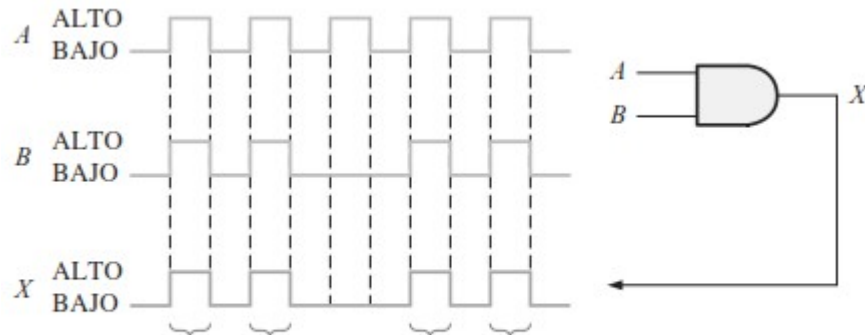
Tablas de verdad

x	y	$x \cdot y$	x	y	$x + y$	x	x'
0	0	0	0	0	0	0	1
0	1	0	0	1	1	1	0
1	0	0	1	0	1		
1	1	1	1	1	1		

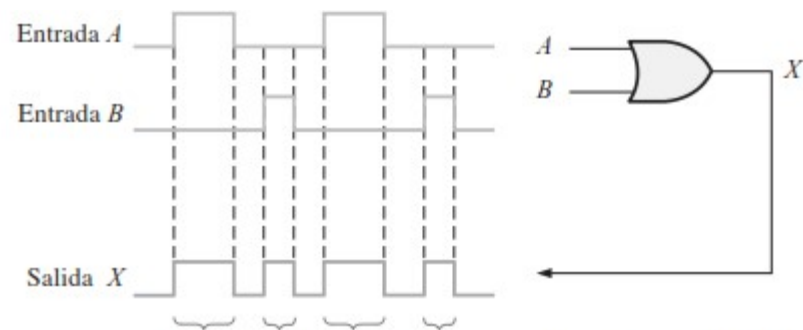
x	y	z	$y + z$	$x \cdot (y + z)$	$x \cdot y$	$x \cdot z$	$(x \cdot y) + (x \cdot z)$
0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0
1	0	0	0	0	0	0	0
1	0	1	1	1	0	1	1
1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1



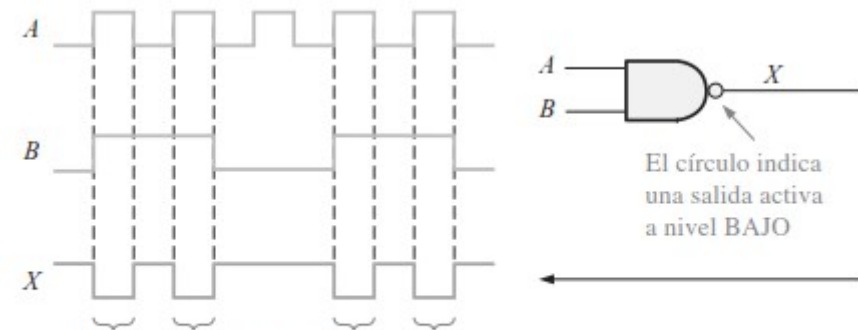
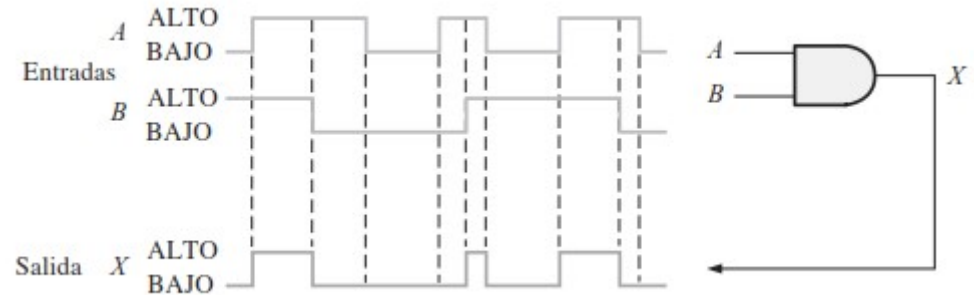
Diagramas Temporales



A y B están a nivel ALTO durante estos cuatro intervalos de tiempo. Por tanto, X está a nivel ALTO.



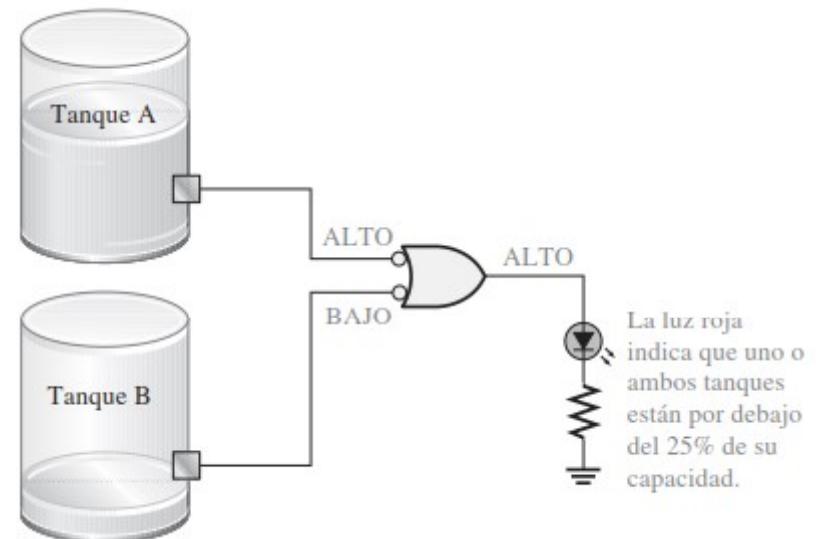
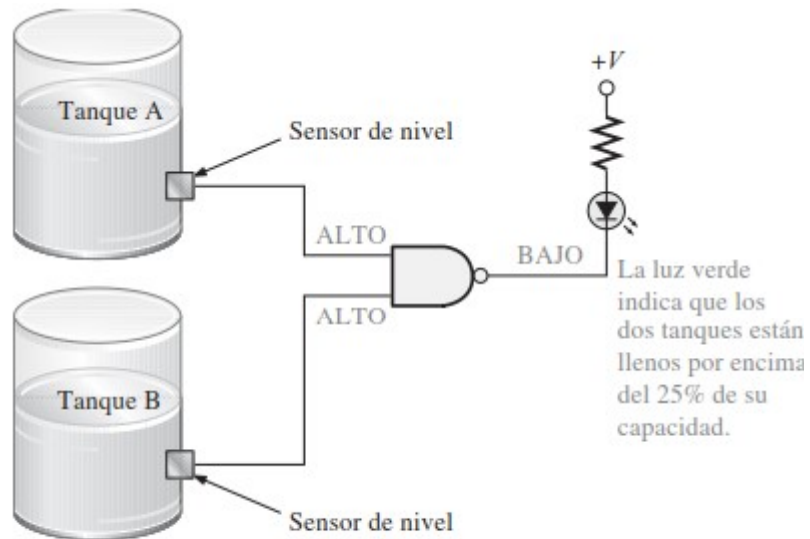
Cuando cualquiera de las entradas o ambas están a nivel ALTO, la salida es un nivel ALTO.



A y B están a nivel ALTO durante estos cuatro intervalos de tiempo. Por tanto, X es un nivel BAJO.



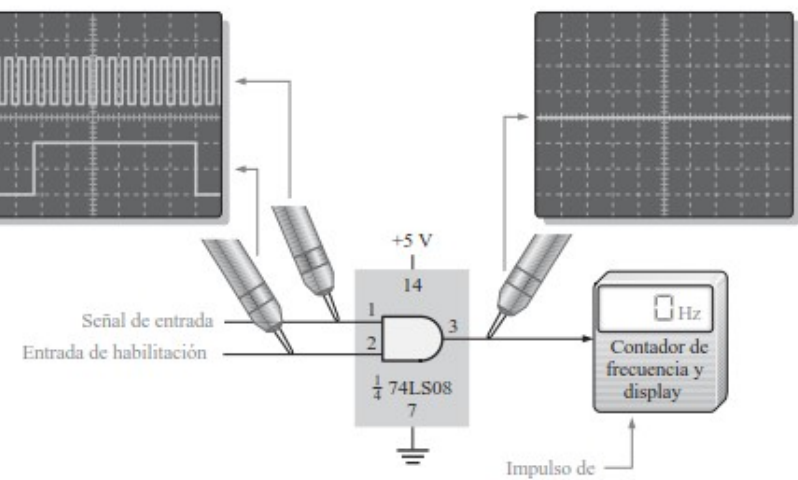
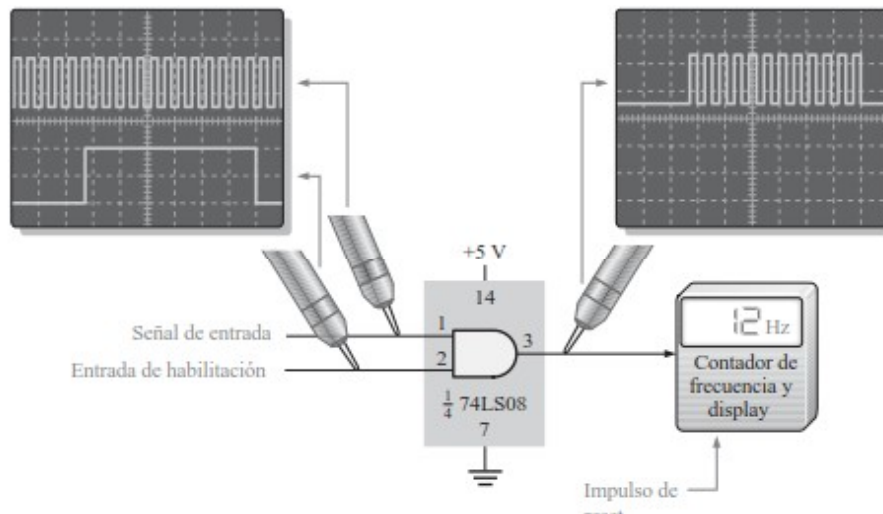
Aplicaciones





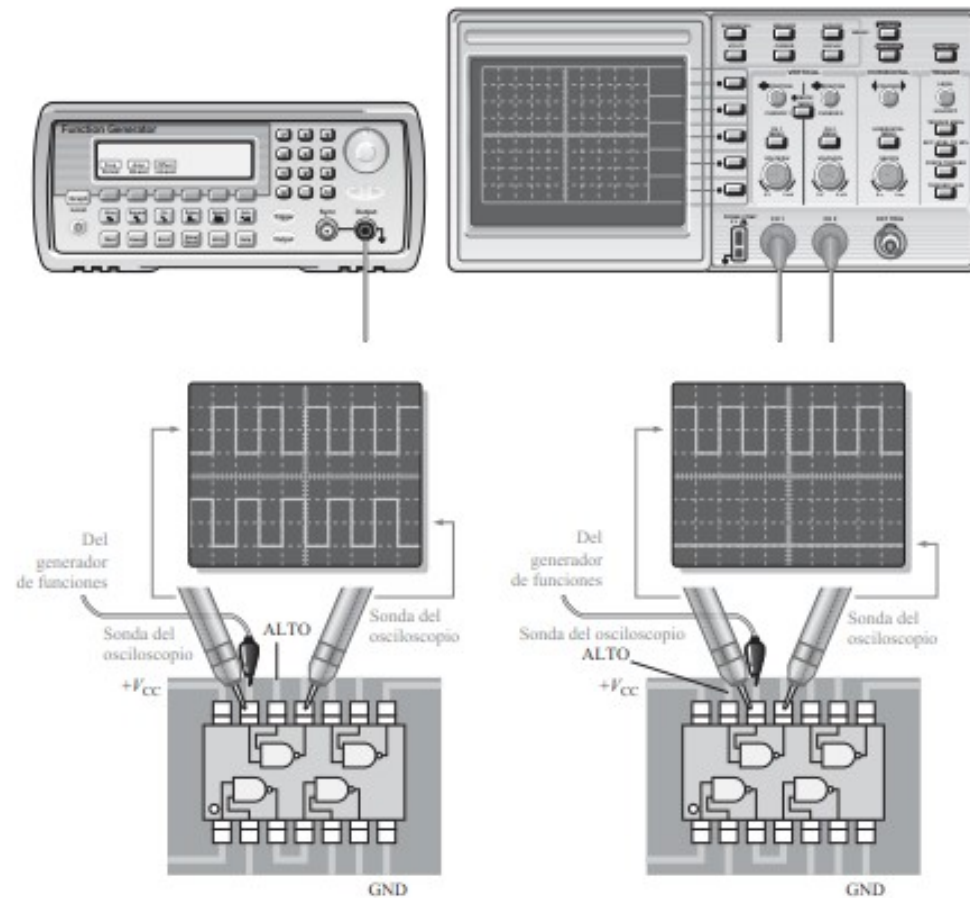
Conexiones y mediciones

Prueba y detección de fallos



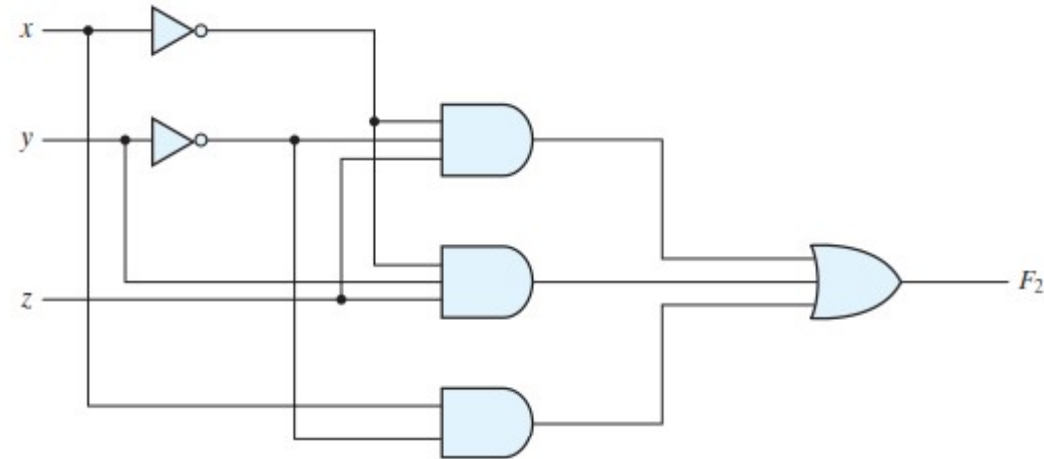


Conexiones y mediciones

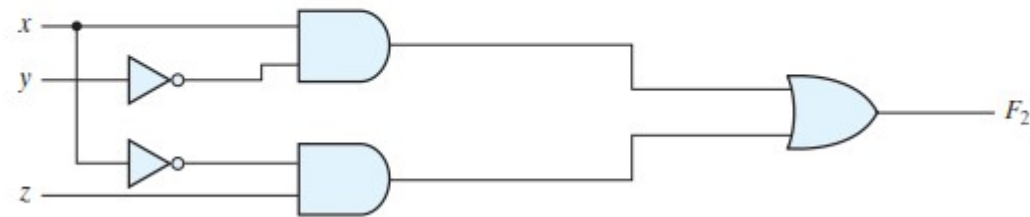




Circuitos a realizar en Simulador CedarLogic



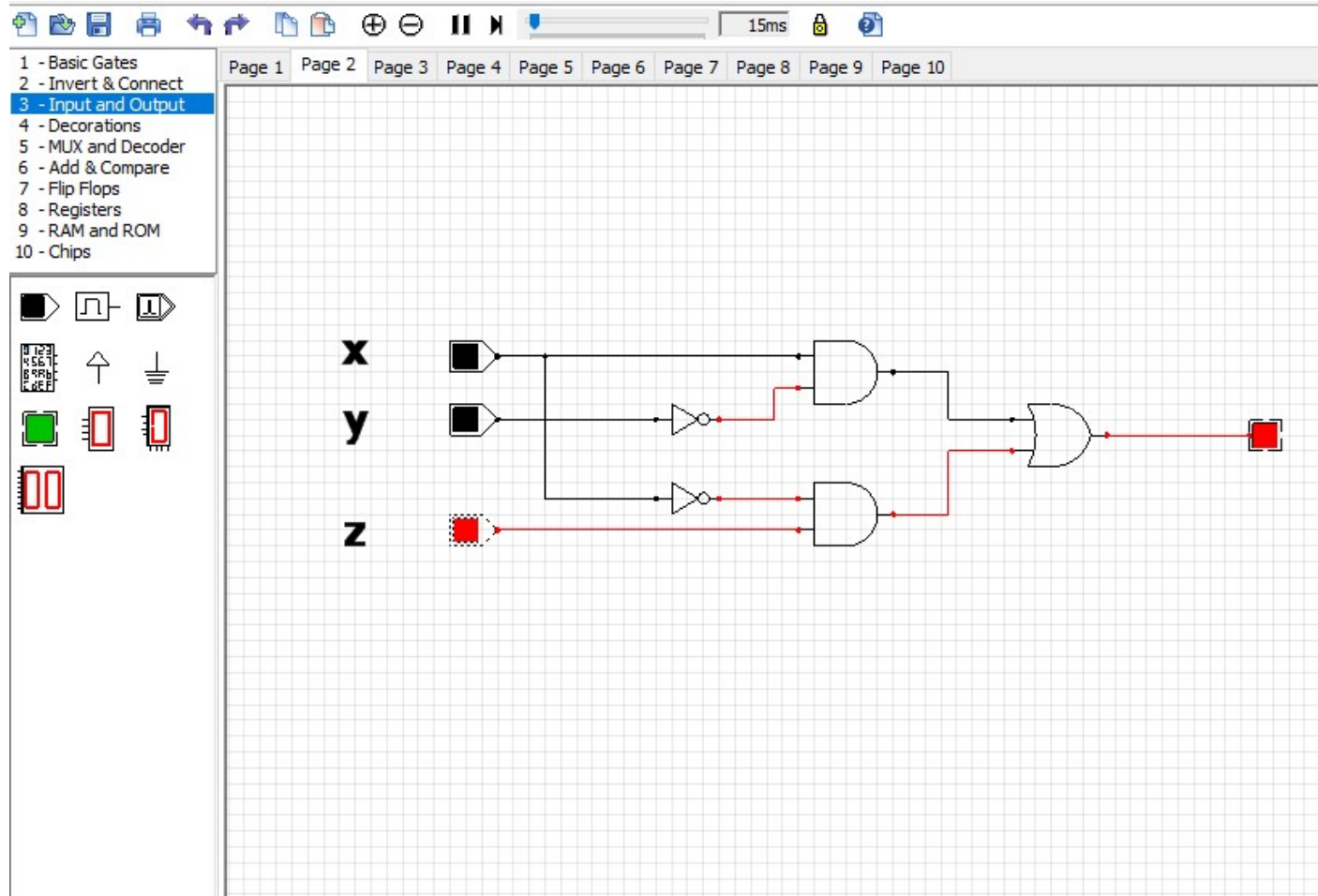
(a) $F_2 = x'y'z + x'yz + xy'$



(b) $F_2 = xy' + x'z$



Simulación del circuito





Algebra de Boole

POSTULADOS DEL ALGEBRA DE BOOLE

$$\begin{array}{lcl} A + 1 = 1 & \longleftrightarrow & A \cdot 0 = 0 \\ A + 0 = A & \longleftrightarrow & A \cdot 1 = A \\ A + \bar{A} = 1 & \longleftrightarrow & A \cdot \bar{A} = 0 \\ \bar{\bar{A}} = A & & \end{array} \quad \begin{array}{l} \text{(DUALIDAD)} \\ \text{---} \end{array}$$

PROPIEDAD CONMUTATIVA

$$A + B = B + A$$

$$A \cdot B = B \cdot A$$

PROPIEDAD DE INVARIANCIA

$$A + 0 = A$$

$$A \cdot 1 = A$$

PROPIEDAD DISTRIBUTIVA

$$A(B + C) = AB + AC$$

$$A + BC \dots N = (A+B)(A+C) \dots (A+N)$$

PROPIEDAD DEL COMPLEMENTO

$$A + \bar{A} = 1 = U$$

$$A \cdot \bar{A} = 0$$



Algebra de Boole

TEOREMAS DEL ALGEBRA DE BOOLE

PRINCIPIO DE DUALIDAD: SE OBTIENE UNA EXPRESION DUAL SI:
(COMPROBAR EN POSTULADOS) $(+) \longleftrightarrow (.)$ $(0) \longleftrightarrow (1)$

ABSORCION

$$A + ABCD\dots N = A$$

$$A \cdot (A+B+C+\dots N) = A$$

ASOCIATIVA

$$A + (B+C) = (A + B) + C$$

$$A \cdot B \cdot C = (A \cdot B) \cdot C$$

NEGACION

$$\overline{\overline{A}} = A$$

$$\overline{0} = 1, \quad \overline{1} = 0$$

DE MORGAN

$$\overline{A+B} = \overline{A} \cdot \overline{B}$$

$$\overline{A \cdot B} = \overline{A} + \overline{B}$$



Fin

De esta parte!!